

OMNIINTELOS S.A.

Niamey DC1 Technical Whitepaper

Sovereign AI infrastructure for the Sahel

Facility	OmniIntelOS Niamey DC1
Design tier	Uptime Institute Tier III (concurrently maintainable)
Commissioned	2024-07
Design IT load	1.6 MW
Author	Cloud & Data Centre Operations

OmniIntelOS S.A. is a fictional company. This document is generated for IntelAI demonstration and evaluation purposes.

1. Rationale

Regional customers in banking, public administration and telecommunications increasingly face data residency obligations that cannot be met by hosting outside the UEMOA zone. DC1 was built to remove that constraint, and to give OmniiIntelOS control over the GPU capacity its computer-vision and NLP workloads depend on rather than renting it at spot prices abroad.

2. Design

Location	Niamey, Niger - dedicated 33 kV feed
Design IT load	1.6 MW across four halls
Redundancy	N+1 UPS, N+1 CRAC, 2N utility/generator transfer
Cooling	Indirect adiabatic, elevated ASHRAE A3 envelope for the Sahel climate
Design PUE	1.38 annualised (measured 1.41 in first full year)
GPU capacity	224 accelerators across training and inference pools
Connectivity	Three carriers, diverse physical entry, 400 Gbps aggregate
Certifications	ISO 27001 (2024), ISO 14001 (2026), PCI-DSS scoped zone

3. Climate and power reality

Ambient temperatures above 45 C for sustained periods and a grid with historically variable availability drove two decisions that differ from a temperate-climate design: an elevated thermal envelope operated to ASHRAE A3 rather than A1, which trades a small amount of hardware lifetime for a materially lower cooling load, and on-site generation sized for full IT load rather than for a short bridge to utility restoration.

The Scope 1 emissions in the ESG series step up visibly from 2024 onward for exactly this reason - generator runtime is a real, measured cost of operating in this location, and the sustainability roadmap addresses it with solar plus storage rather than by excluding it from the boundary.

4. Measured operating data

Period	Energy (kWh)	Renewable %	Scope 2 (tCO2e)	Uptime %	P99 (ms)
2024-07	156,257	41.3	56.9	99.995	193
2024-09	192,892	41.8	69.6	99.995	210
2024-11	230,473	43.4	80.9	99.995	200
2025-01	235,190	42.0	84.5	99.995	196
2025-03	240,114	45.9	80.6	99.762	203
2025-05	245,181	45.2	83.2	99.776	184
2025-07	251,092	47.8	81.3	99.794	201
2025-09	257,464	60.5	63.1	99.995	190
2025-11	264,806	62.4	61.7	99.995	194
2026-01	272,066	62.2	63.7	99.995	181
2026-03	280,044	66.6	58.0	99.995	177
2026-05	288,285	65.2	62.2	99.995	178

5. Roadmap

- DC2 Dakar - second availability zone, target commissioning 2027, enabling active/active regional failover.
- Solar plus battery storage phase 2, lifting renewable share past 70% and reducing generator runtime.
- Liquid cooling pilot for the training pool as accelerator density rises.
- Carbon-aware scheduling: shifting non-urgent training to high-renewable windows.